Progress Report

1. Basic Details

- **Project Title: AI Integrated Machine vision technology for equipment condition monitoring and feed control in Indian Industries with a use case from our region.**

- **Name of IIT: IIT BHILAI**
- **Project Mentor:** Dr. Jose Immanuel
- **Team Lead Name: Rajana Venkata Mohit**
- **Team members' Name: Nagulapalli Shavaneeth Gourav, Pulidindi Maneesh Bhaskar**

- **Project Start Date:** 06/05/2024
- **Expected Completion Date:** 24/07/2025

2. Concept Note

A physical survey of industries in Chhattisgarh revealed that **efficient separation of mixed input materials in production lines remains a critical challenge**, directly affecting productivity, efficiency, and product quality. Traditional manual sorting is labour-intensive and inconsistent, while existing automated systems lack adaptability for diverse materials, leading to inefficiencies.

A particularly pressing issue was observed in **the recycling industry**, where tangled wire inputs frequently **choke processing units**, disrupting downstream operations. The challenge extends beyond decluttering; it involves **predicting when intervention is needed** and ensuring continuous, adaptive material feed to prevent equipment bottlenecks.

Proposed Solution

We propose a **camera-based feed monitoring system** integrated with a robotic arm for **real-time decluttering and adaptive material feed control**. This system will:

- **Monitor and analyze input materials** using computer vision.

- **Trigger robotic intervention** only when necessary, ensuring efficiency.

- **Optimize downstream feed rates** using condition monitoring, reducing system choking.

While this project focuses on recycling, the developed **system and protocols** can be scaled to multiple industries facing similar challenges, such as manufacturing and agriculture. Beyond solving an immediate industry problem, **this project lays the foundation for a technology startup** that can cater to industries nationwide.

3. Initial Stages & Planning

- **Literature Review & Benchmarking:**

  we have conducted an extensive review of existing research and technologies in image processing, feature extraction, and automated material separation. Our focus has been on **Convolutional Neural Networks (CNNs)** for object detection and classification, particularly in industrial settings. Key areas of study include:

  State-of-the-art CNN-LSTM architectures (e.g., ResNet, EfficientNet, YOLO) for real-time image processing.

  Feature extraction techniques to distinguish materials based on shape, texture, and colour.

  Challenges in industrial applications, such as handling non-standard object shapes and high-speed processing constraints.

  ### Our benchmarking process included:

  Industry Practices: We studied manual sorting methods and existing automated systems, identifying inefficiencies in speed, accuracy, and adaptability.

  Technological Comparison: We compared various image-processing techniques and CNN models to determine their effectiveness in real-time industrial environments.

  Performance Metrics: We assessed factors such as model accuracy, processing speed, and computational efficiency to establish a baseline for improvement.

- **Project Design & Methodology:**

Approach for Material Separation and Adaptive Feed Control

1. **Problem Understanding & Data Collection**

    - Conduct **physical surveys** to analyze the production workflow.
    - Identify **choking points** in the material separation and feeding system.
    - Collect real-world **image data (RGB, POINT CLOUD)** using LIDAR (intel I515) mounted along the conveyor belt.
    - Gather **current sensor data** from the hammer mill to monitor performance parameters.

2. **Computer Vision for Material Classification & Separation**

    - Implement a **CNN-based image processing model** to classify materials based on their shape, size, and texture.
    - Extract **features from mesh data** for better material distinction.
    - Detect **clogging patterns** in the conveyor system and trigger the **robotic decluttering mechanism**.

3. **Adaptive Feed Rate Optimization**

    - Develop a **jamming prediction model** using sensor readings and image analysis.
    - Implement a **feedback control loop** to regulate material input dynamically.
    - Use an **electromagnetic conveyor system** to modulate feed rate based on real-time analysis.

4. **System Integration & Deployment**

    - Integrate the **computer vision model** with industrial control systems.
    - Implement a **real-time decision-making system** for feed control.
    - Conduct **pilot testing** at the industry site to measure efficiency improvements.

5. **Future Scope & Commercialization**

    - **Optimize the AI model** for real-time performance improvements.
    - **Expand the application** to other industries requiring material separation.
    - Develop a **market-ready product** to launch as a technology startup.

- **Resources & Infrastructure:** GPU, Lidar, 3D printer, Fusion 360, GitHub, Intel RealSense.

4. Journey & Implementation

- **Fund Utilization**

Sl. No.	Budget head	Amount
1.	Consumables	50,000
2.	Contingent expenditure	6,500
3.	Travel expenditure	6,000
	Total	**62,500**

- **Milestones Achieved:** List of completed tasks with timelines.

Camera Setup

- 3D printed the camera and jetson holders
- Made a wooden box
- Kept a fan for ventilation purpose in the box

Data Collection

- Successfully collecting the current data using current sensor data
- Successfully collecting the RGB image, Depth image, Point cloud of the data

AWS Uploading

- SSH connection for controlling the jetson
- Automated jetson and camera to upload the data directly into AWS

ARM design

- **Challenges Faced**

SSH Control

Optimization of the camera parameters

Position and orientation of the camera to get optimized data

Noise Removal from the data

Model Training

5. Final Output & Results

Key Findings

- We observed a Spike in current just at the time of clogging.
- We also observed current is increasing when the material going into the mill is increasing.

Impact of the Project

1. **Enhanced Efficiency** – Automating material separation and adaptive feed control reduces manual labor, improves consistency, and minimizes downtime.
2. **Increased Productivity** – Intelligent feed monitoring ensures optimal machine utilization, leading to higher throughput.
3. **Cost Savings** – Reduces operational costs by minimizing human intervention and preventing equipment damage due to choking.
4. **Scalability** – The AI-driven system can be adapted across multiple industries facing similar material-handling challenges.
5. **Sustainability** – Improved material sorting leads to better recycling efficiency, reducing industrial waste

Applications of the Project

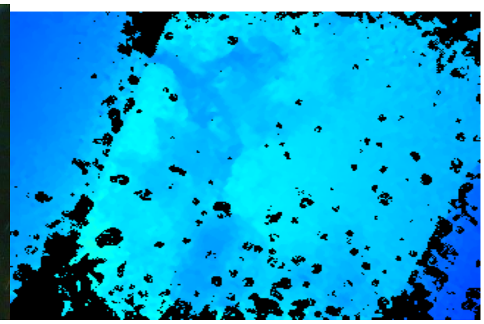
1. **Recycling Industry** – Sorting mixed materials (e.g., tangled wires, plastics, metals) efficiently.
2. **Manufacturing** – Optimizing raw material feed and preventing machinery clogging.

6. Share a pictures/ videos of the prototype

Camera Setup:

Collected Data

RGB Image



Depth Image

Acknowledgement of Receipt of Hyundai Hope Scholarship

I, RAJANA VENKATA MOHIT, along with my teammates NAGULLAPALLI SHAVANEETH GOURAV, PULIDINDI MANEESH BHASKHAR, hereby acknowledge that I have received the full amount of the **Hyundai Hope Scholarship** under the **IIT Innovators category**.

- **Total Amount Received:** ₹62,500

I confirm that the total scholarship amount has been successfully credited to my registered bank account.

I further acknowledge that I have completed all formalities required for receiving the scholarship and understand that this concludes the disbursement process.

Signature: R.V.Mohit
Date: 24-03-2025
Contact Number: 9014168687

Thank you for this opportunity and support.